Nathan Brewer

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Education

Swarthmore College Swarthmore, PA
Bachelor of Science in Engineering, Minor in Applied Mathematics *Expected May 2028*

- Major GPA: 3.96, Cumulative GPA: 3.89
- Relevant Coursework: Mechanics of Solids, Thermofluid Mechanics, Linear Physical Systems Analysis, Principles of Computer Engineering, Electrical Circuit Analysis
- Honors and Memberships: Sigma Xi, American Society of Mechanical Engineers

Work Experience

Research and Development Intern May 2026 – August 2026
Ryzing Technologies *Staunton, VA*

- Led three design iterations of tent circumferential straps for US Army Arctic Medical Operations, increasing tensile strength and standardizing textile dimensions for manufacturing; personally sewed production units
- Validated roll MOI of USMC vehicle model to within 1% of datasheet specifications using MATLAB analysis and lead-weight ballasting, enabling wave pool roll testing
- Automated a manual, cable-actuated wind tunnel throttle with a servo-driven, PID-controlled system running programmable RPM profiles from CSV, enabling repeatable, client-customizable wind testing
- Developed supplier evaluation matrices and coordinated directly with vendor representatives to source EMI shielding fabric for a secure communications program
- Produced dimensioned drawing packages with tolerances and datum references across all design projects

Engineering Research Associate October 2024 – Present
Swarthmore College *Swarthmore, PA*

- Built a biologically accurate robotic barnacle nauplius model, enabling University of Washington collaborators to study the effects of morphological variation on hydrodynamic performance
- Designed a fully modular drive assembly with three appendage levels, allowing investigation of vertical positioning of appendage sets
- Developed a light-activated motor control circuit with adjustable threshold and drive voltage, providing locomotion across a range of speeds and ambient lighting conditions
- Consolidated actuation from three independent drive motors to a single common drive shaft, reducing mechanical complexity and power draw while maintaing accurate motion
- Reduced mechanism height by 20% relative to prior designs, improving profile accuracy for fluid testing

Physics Teaching Assistant September 2025 – December 2025
Swarthmore College *Swarthmore, PA*

- Selected to lead weekly help sessions on mechanics topics (kinematics, dynamics, energy/momentum) for 60+ students in General Physics I

Extracurricular Activities

Swarthmore Robotics

- Designed and fabricated weapon, armor, and control/drive systems of 1 lb. BattleBot
- Use Fusion360 and ANSYS to run dynamic impact simulations for weapon and armor designs
- Designed modular antweight class BattleBots to introduce new members to design principles
- Leading development and fabrication of NHRL-compliant Beetleweight (3 lb.) combat robot

Model Rocketry

- Designed a custom layup and curing setup to hand-wrap a carbon fiber body tube
- Model rocket designs in SolidWorks and OpenRocket to determine stability and flight characteristics
- Manually machined ram-set tooling to create custom bentonite clay nozzle geometry
- Achieved roughly 20% burn rate increase in home-cast KnSu rocket motors through iron oxide doping

Skills

Design: SolidWorks, Fusion360, Onshape, ANSYS Mechanical, KiCad, MultiSim
Computer Languages: MATLAB, Python, SQL
Fabrication: MIG/TIG welding, manual lathe, manual mill, basic carpentry, 3D printing, soldering